WQF7004
Data Analytics in Artificial Intelligence

Data Mining and Acquisition

Credit to : IE:4172 Big Data Analytics Stephen Baek

Sea of Information

- Internet data are extremely prevalent
- They can be useful in many applications:
  - Predicting outcomes of political elections
  - Market trend research
  - Sentiment/reputation analysis
  - Stock market prediction
  - Sports science
  - Diffusion of information
  - Natural disasters
  - Diseases, epidemiology, public health
  - … the list goes on and on

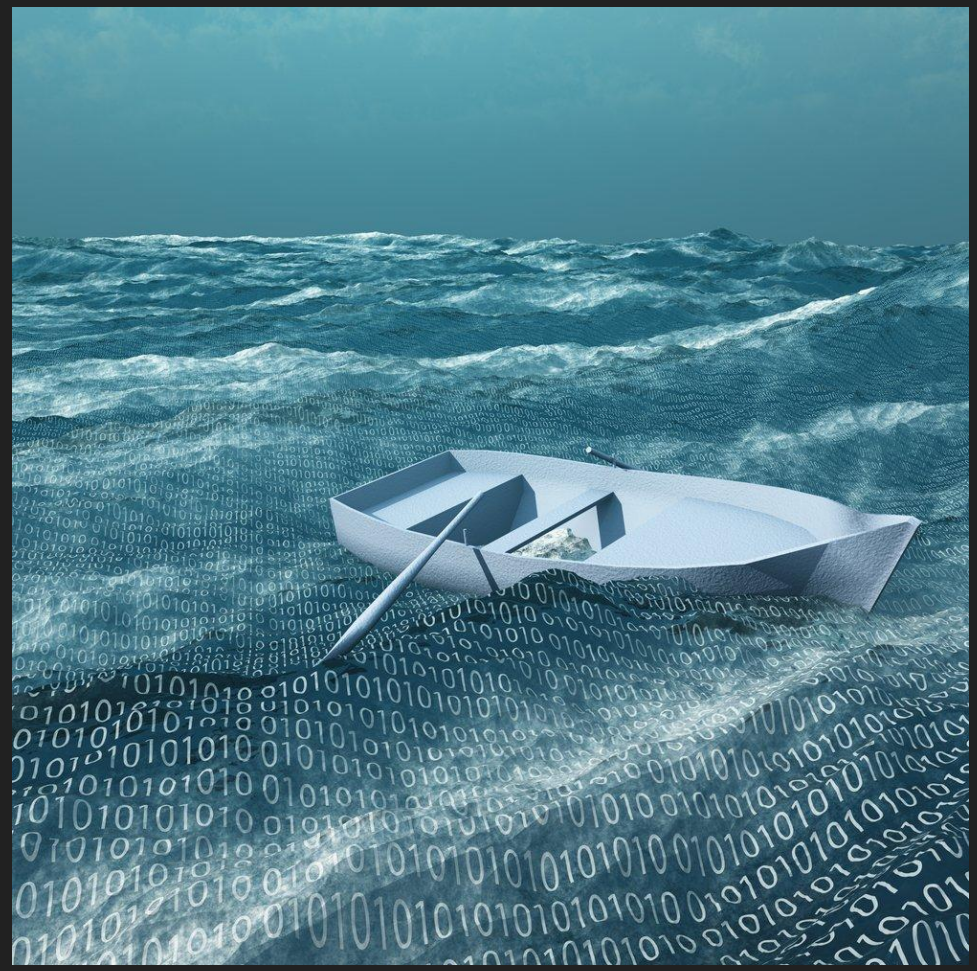


Image Source: Unknown

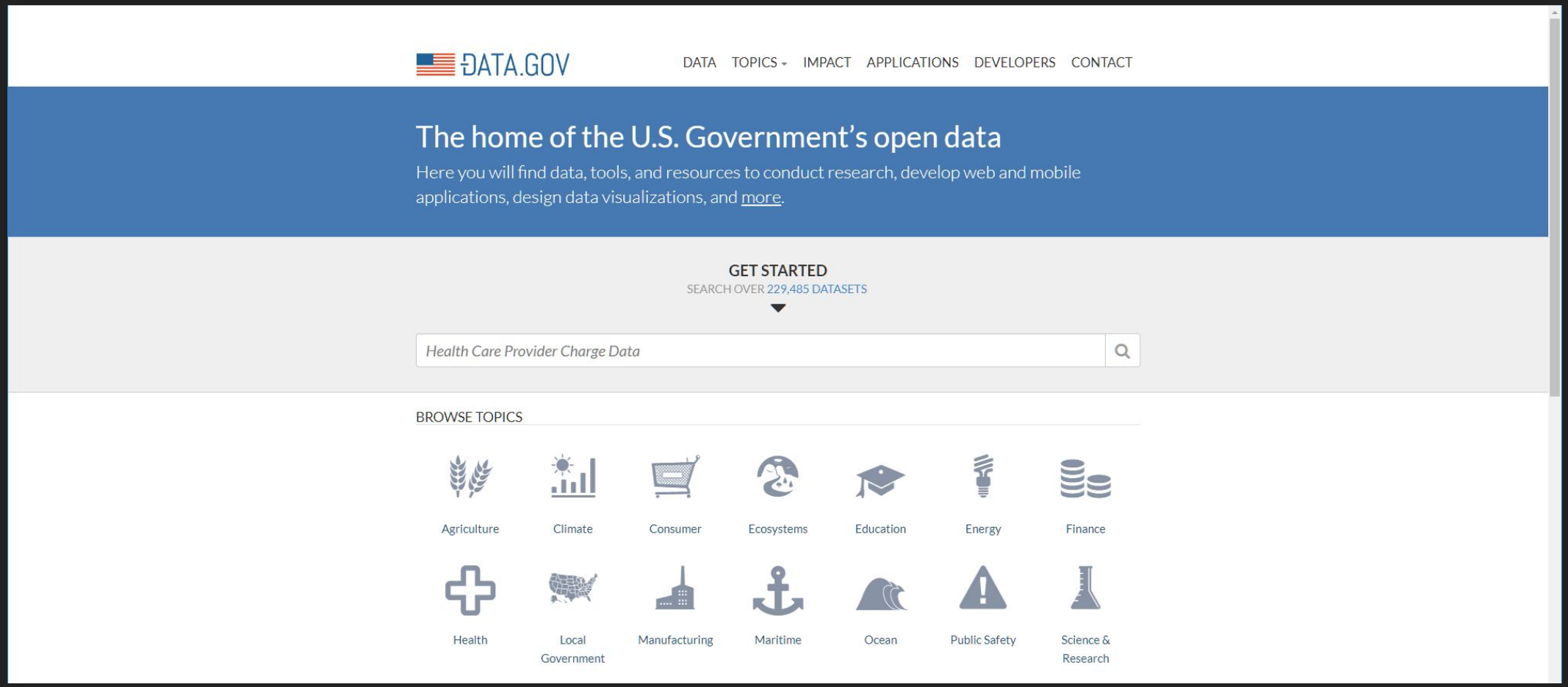
Data is the new oil

- We have to “mine” it…
  - Publicly available datasets
    - Raw files made available for download
    - e.g. UCI ML repository, Kaggle competitions, data.gov, NIH Chest X-ray Dataset, …
  - Web crawling/scraping
    - Automated bots/macros to collect data from the web
    - Navigate through websites by tracking down the links
    - e.g. Search engines!
  - API - Application Programming Interface
    - A programing interface to send query & retrieve data
    - e.g. Twitter API
  - Proprietary datasets

Image Source: Wikipedia

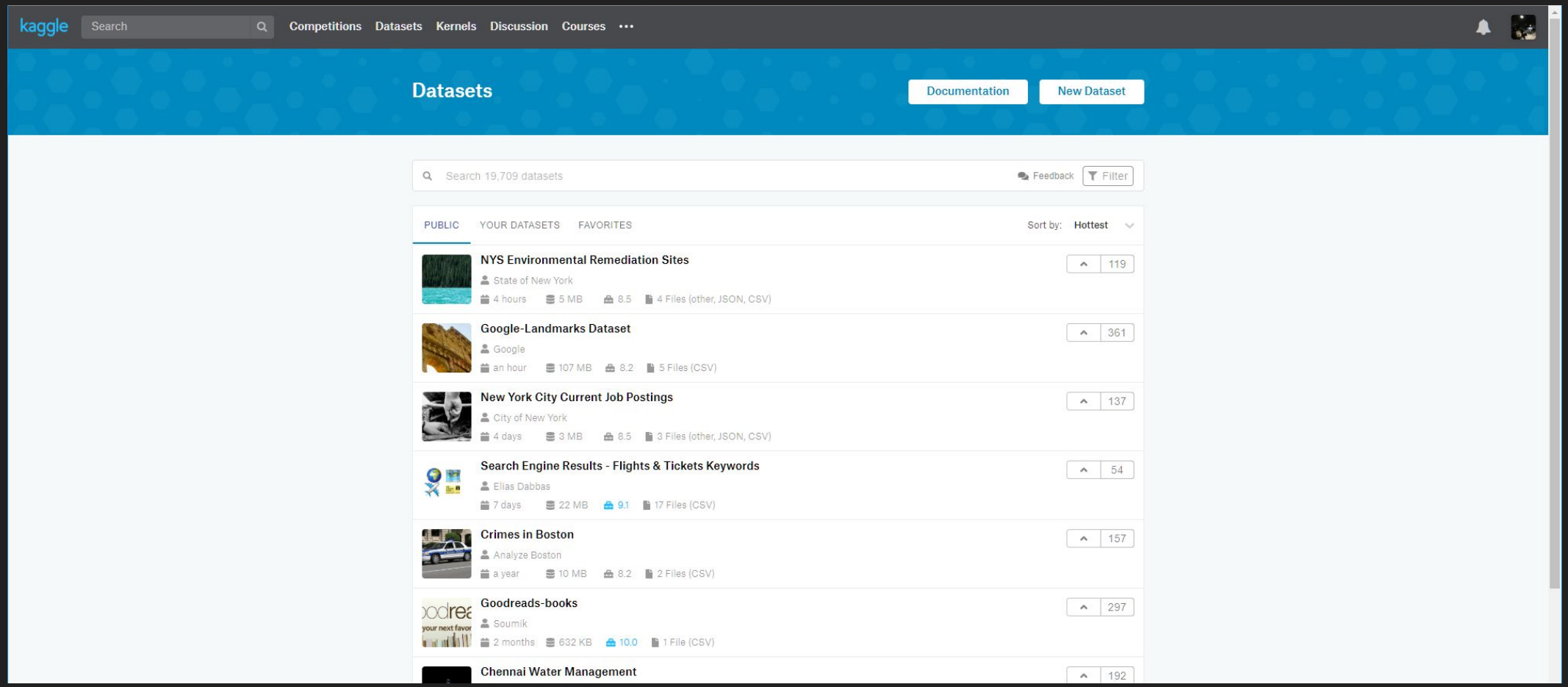
Public Datasets

- https://www.data.gov/



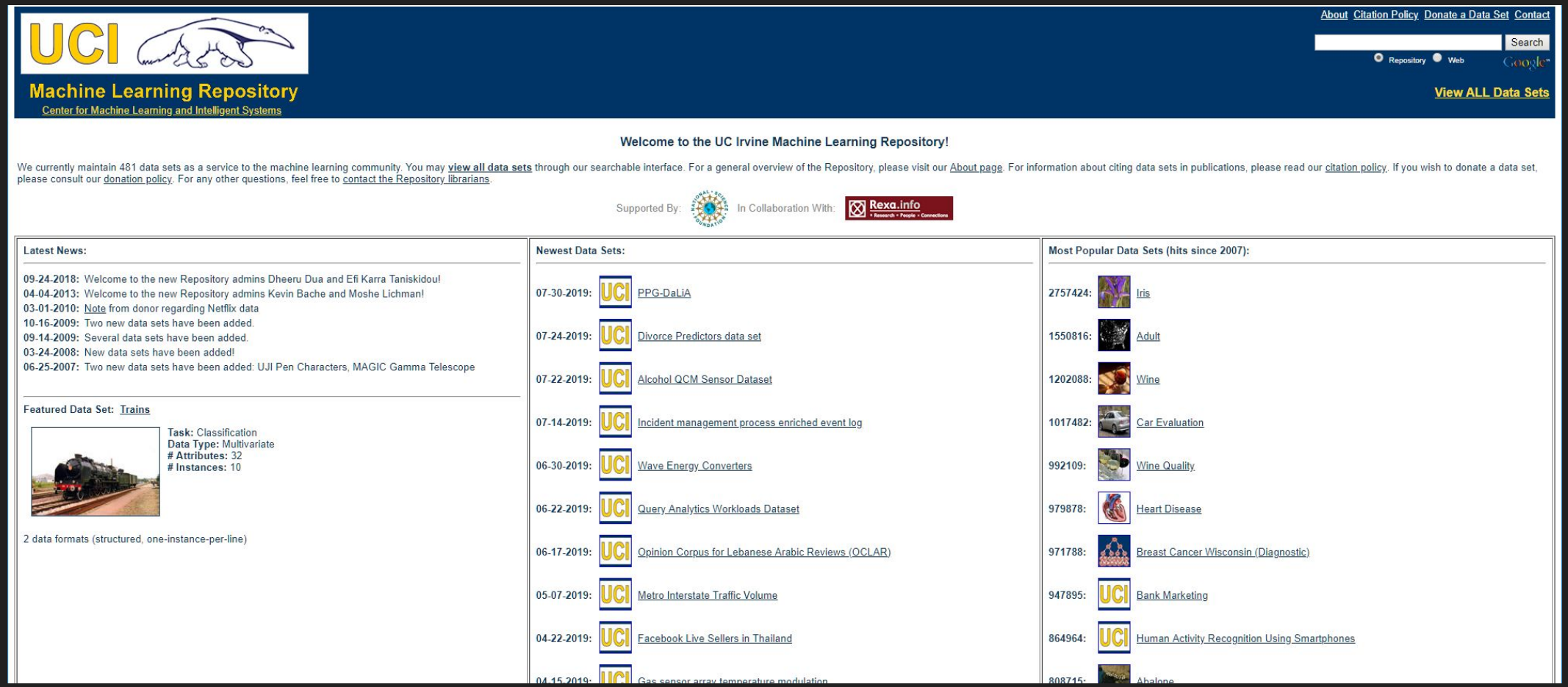
Public Datasets

- https://www.kaggle.com



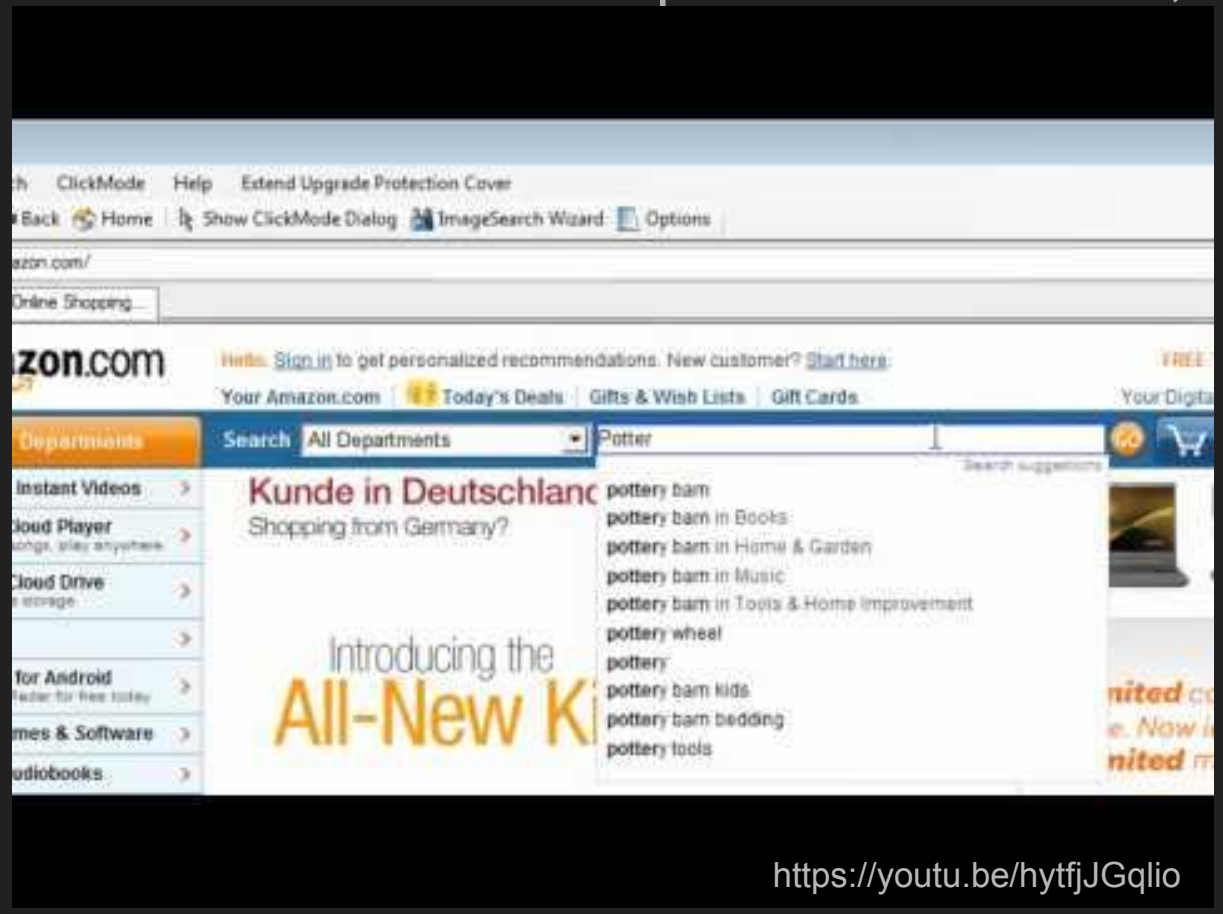
Public Datasets

- https://archive.ics.uci.edu/ml/index.php



Web Crawling & Scraping

- Data mining from websites can be incredibly tedious and repetitious
- Web browser macros can automate repetitive web clicks, filling in forms, etc.



https://youtu.be/hytfjJGqlio

Web Crawling & Scraping

- Crawler: aka web robot, or web spider
  - A software program that automatically traverses hyperlinks
  - Systematically browses the world wide web
  - Examples:
    - Googlebot: collects documents from the web to build a searchable index.
    - Xenon: is a web crawler used by government tax authorities to detect fraud

- There are many open source crawlers:
  - For example: https://github.com/scrapinghub
  - BeautifulSoup, LXML

Web Crawler Policies

- The behavior of a web crawler is the outcome of a combination of policies
  - a *selection policy* which states the pages to download,
  - a *re-visit policy* which states when to check for changes to the pages,
  - a *politeness policy* that states how to avoid overloading Web sites.
  - a *parallelization policy* that states how to coordinate distributed web crawlers.

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  - a *parallelization policy* that states how to coordinate distributed web crawlers.
- Web crawlers are not always welcome
  - A not so well-behaved crawler can be blacklisted
  - robot.txt: a special file located on a web server that enforces restrictions
    - ‘Allow’ tag: list of pages that can be accessed
    - ‘Disallow’ tag: list of pages that should not be indexed
  - HTML META tags: does the similar thing with robot.txt
    - <META name=”ROBOT” content=”NOFOLLOW”>
    - <META name=”GOOGLEBOT” content=”NOINDEX”>

Application Programming Interface (API)

- Set of functions, routines, protocols, and tools for building software applications
- APIs define the standard way of accessing data
- Examples:
  - Twitter API: https://dev.twitter.com
  - Facebook API: https://developers.facebook.com
  - Yahoo! Finance API
  - Google Map API
  - …

Reading Topic 1

- David Balley. (2018). Why outliers are good for science?
  - https://rss.onlinelibrary.wiley.com/doi/pdf/10.1111/j.1740-9713.2018.01105.x

- Discuss:
  - What is the Gaussian distribution (the bell curve) and what is the Cauchy distribution?
  - Is real-world measurement closer to the Gaussian or Cauchy? Why do you think is the reason?
  - What's the criteria commonly used to determine outliers? How can they be wrong?
  - What is the author's point to claim that outliers might actually be good for science?

Reading Topic 2

- Candace Corbeil - Gaps in the Spreadsheet
  - https://www.apa.org/science/about/psa/2016/02/gaps-spreadsheet
- Gerhard Svolba - The origin, detection, treatment and consequences of missing values in analytics.
  - http://analytics-magazine.org/missing-values/

- Discuss:
  - What are the three types of missing data?
  - What is multiple imputation how can they be useful for data that are missing at random?
  - In case of systematic (non-random) missing data, would you still use multiple imputation? Or what else can you do?